

# Overview and Importance of Biology in Engineering



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## Introduction

- There are a number of options for undergraduate study in our country that now-a-days seem acceptable to the society and are considered worthy enough to pursue.
- This is a healthy situation and the social acceptance of many undergraduate study options needs to improve further.
- Not so long ago in our country, there were only two socially preferred undergraduate streams for study—**Engineering** and **Medicine**.
- For engineering, mathematics is required and biology is not.
- Irrespective of the group they choose to pursue at the higher secondary level, some students have an innate interest in biology, whereas others do not.
- The students who do not have an interest in biology or are neutral, when exposed to uninspired teaching of biology as a part of science at the high school, most likely develop a hatred toward it.

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## NEED FOR BIOLOGY

- The previous few centuries saw a better fundamental understanding of the physical and chemical world through advances in physics and chemistry.
- The better understanding and advances gave rise to technologies and products, such as computers, communication devices, aircraft, and others that revolutionized life.
- Since this is the century of biology, a similar phenomenon is expected, which will lead to probably another revolution.
- Many engineers are expected to contribute to a biological aspect to fuel this revolution.
- Therefore, the engineering undergraduates need to be suitably exposed atleast to the very minimum biology.
- So that they would atleast be able to consider a biological system/aspect in which they could later make appropriate contributions, through their main expertise, say electrical engineering, mechanical engineering, computer science, materials engineering, or any other.

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## Why do engineers need to study the principles of biology?

- Principles of mechanics applied to understand biological systems.
- Biomechanics – is the application of mechanical principles in the study of living organisms including their kinematics and kinetics, it views human body as a collection of levers, made of bones which are moved by its muscles. In sport, used to analyze the performance level of athletes. Focuses on body segments and its interaction with the surrounding environment.
- Understanding Pharmacokinetics and Pharmacodynamics of medicines.

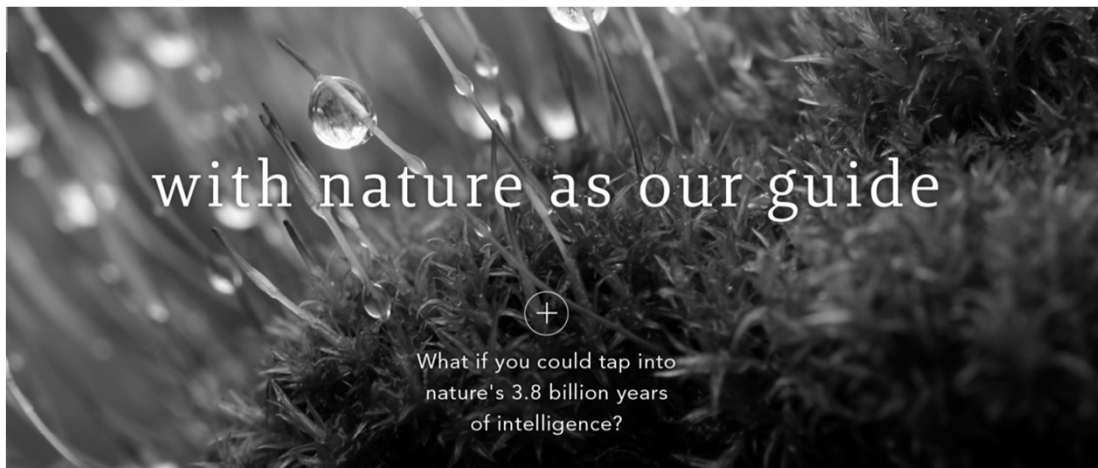
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## Why do engineers need to study the principles of biology? cont.

- Mechanobiology – in the level of cells – it dwells on the behavior of physical forces and transfer in cell and/ or tissues.
- Nanotechnology – carbon nanotubes as drug delivery systems used in cancer therapy
- Computational Fluid Dynamics (CFD) – an engineering tool that connects mechanics to mathematics and software programming to execute simulation performing how a fluid (liquid or gas) flows based on Navier-Stokes equations (used to describe the motion of viscous fluid substances).
- **Bionics** or **biologically inspired engineering** is the application of biological methods and systems found in nature to the study and design of engineering systems and modern technology.

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## Why do engineers need to study the principles of biology? cont.



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## Why do engineers need to study the principles of biology? cont.

**DEFINE**  
Challenge

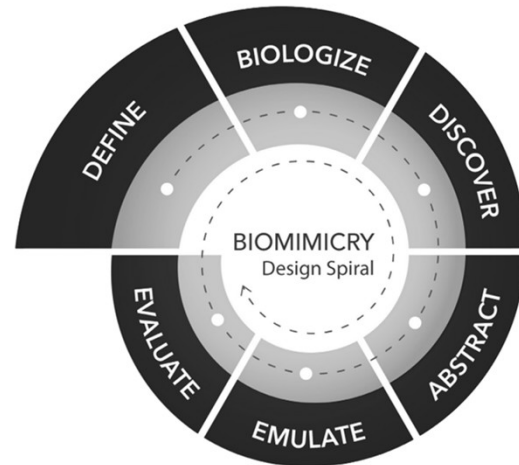
**BIOLOGIZE**  
Function & Context

**DISCOVER**  
Biological Strategies

**ABSTRACT**  
Design Strategies

**EMULATE**  
Nature's Lessons

**EVALUATE**  
Fit and Functionality



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## Why do engineers need to study the principles of biology? cont.

- Biomimicry is an innovative approach that studies nature's time-tested patterns and strategies to solve human challenges sustainably, emulating biological forms, processes, and systems for new designs, products, and policies. It involves looking to nature as a vast library of efficient solutions, like Velcro inspired by burrs to create more resilient and environmentally friendly human-made technologies and systems.
- Synonyms include **biomimetics**, **bio-inspired design**, and **biologically inspired design**.

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## Shinkansen Sonic Boom

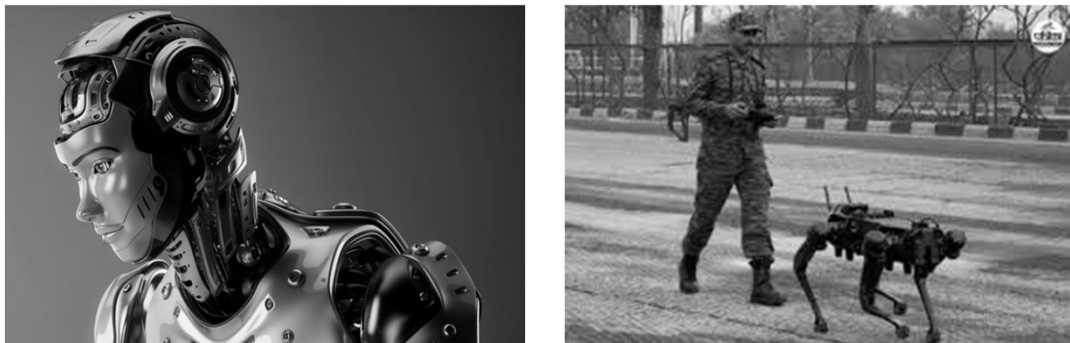
Many man-made things have significant scope for optimising their design. For example, Shinkansen, Japan's high-speed bullet train, plays an important role in Japan with a coverage of close to 3000 km.



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## Bio-robotics

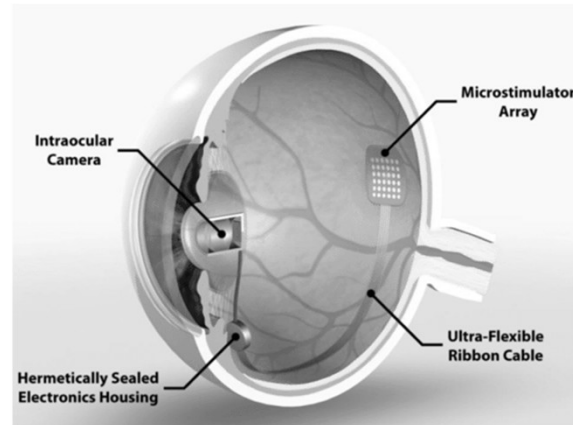
- It refers to robots that are inspired by biological entities or the use of biological components in robots.
- We are possibly familiar with the concept of bio-robotics from many sci-fi movies and TV shows, which show robots with human like features.



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## Retinal Prosthetic

- It is a device that reportedly replaces lost photoreceptor function by transmitting computer-processed video images to an array of electrodes or via light sensors placed in the epiretinal or subretinal space.
- Many engineering fields were effectively harnessed along with biology to provide eye-sight to people who could see due to retinal diseases such as macular degeneration.



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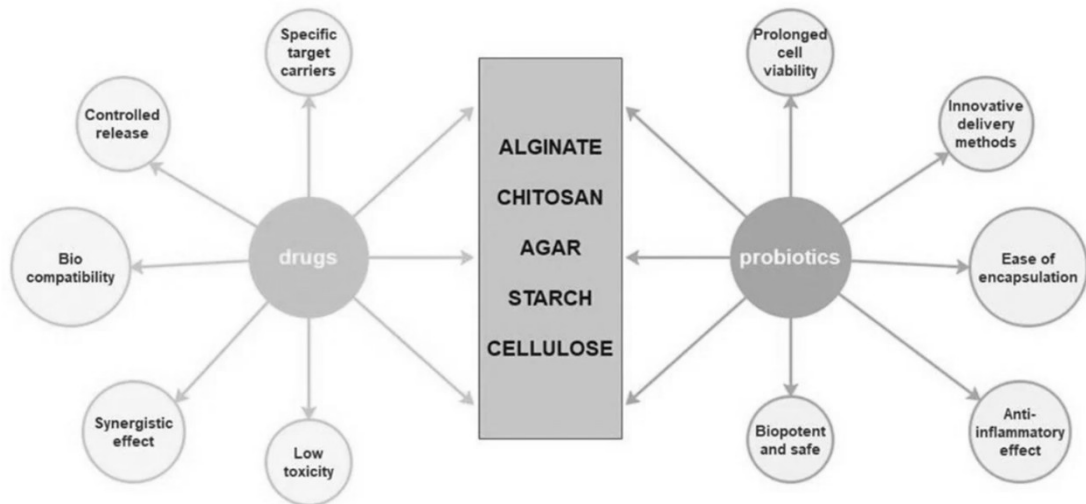
## Biopolymer

- Biopolymers are polymers that are produced by or derived from living organisms, such as plants and microbes, rather than from petroleum, the traditional source of polymers.
- The primary sources of biopolymers are renewable.
- Many biopolymers are biodegradable, but not all. **Polythioesters are non-biodegradable** by microorganisms, which represent a novel non-biodegradable bioplastic material.
- Biopolymers consist of monomeric units that are covalently bonded to form larger molecules. There are three main classes of biopolymers, 1. polynucleotides, 2. polypeptides and 3. polysaccharides.

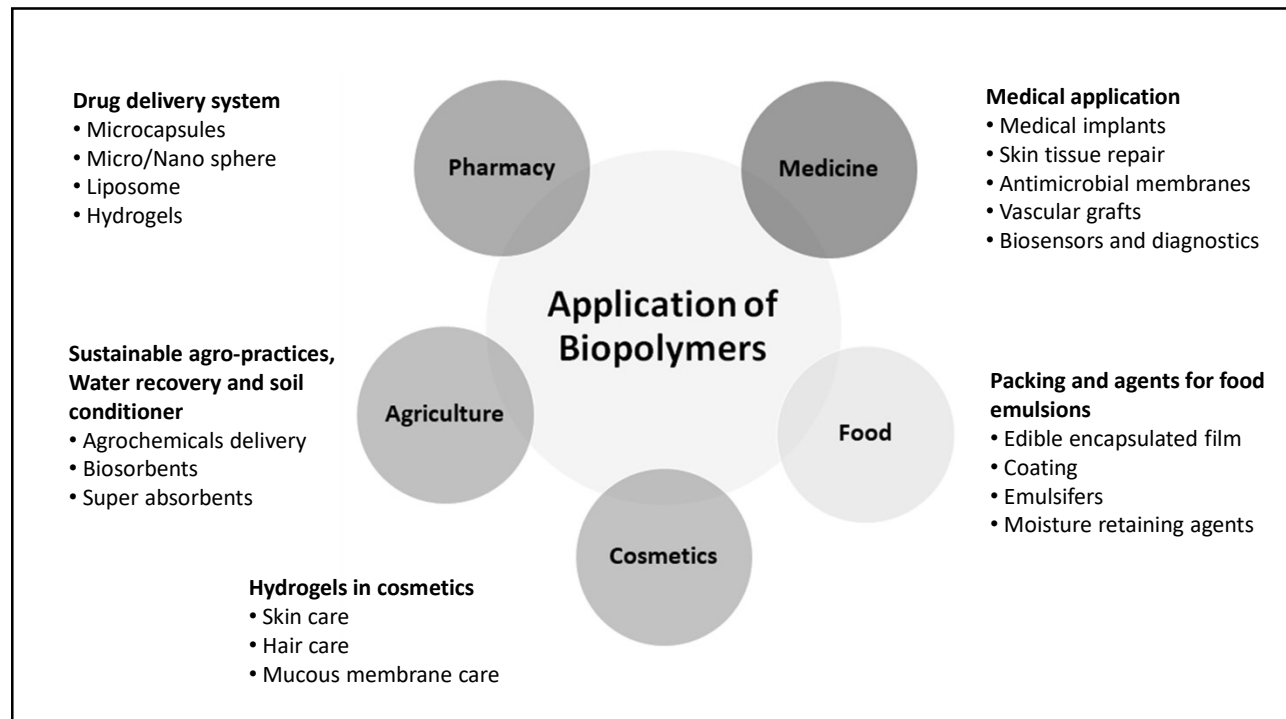
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# Biopolymer

## Biopolymers-based delivery systems



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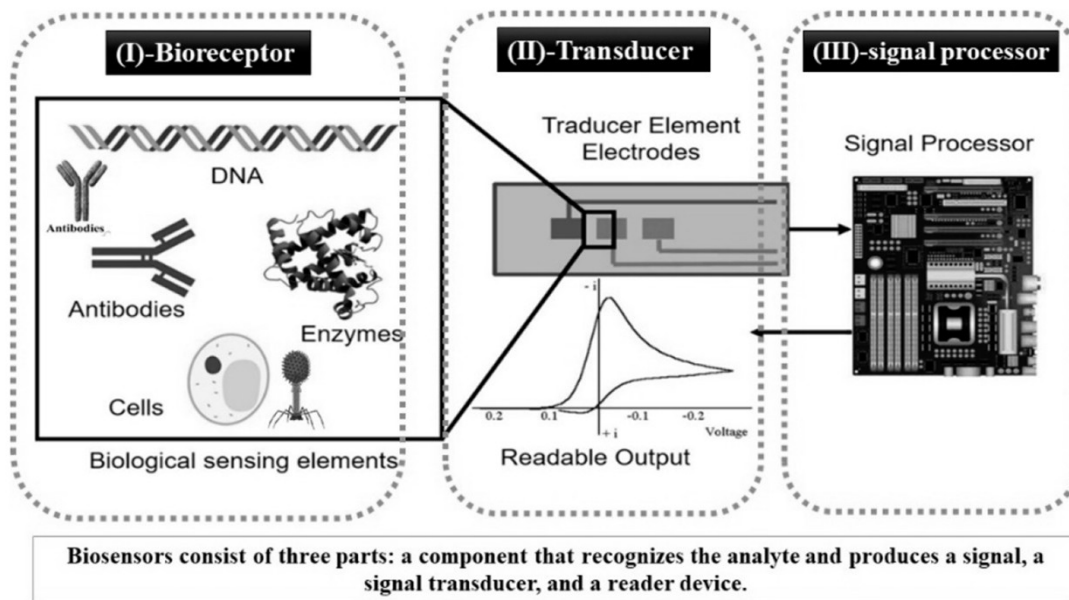
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## Bio-sensors

- These are devices that are used to measure many different parameters such as analyte concentrations. They are used for diverse purposes such as analysis, toxicology, medical diagnosis, environmental monitoring, and others .
- A typically biosensor consists of a bio-receptor (enzyme/antibody/ cell/ nucleic acid/aptamer), transducer component (semi-conducting material/nanomaterial), and electronic system which includes a signal amplifier, processor & display.
- Bio-receptor interaction is measured by the biotransducer which outputs a measurable signal proportional to the presence of the target analyte in the sample.

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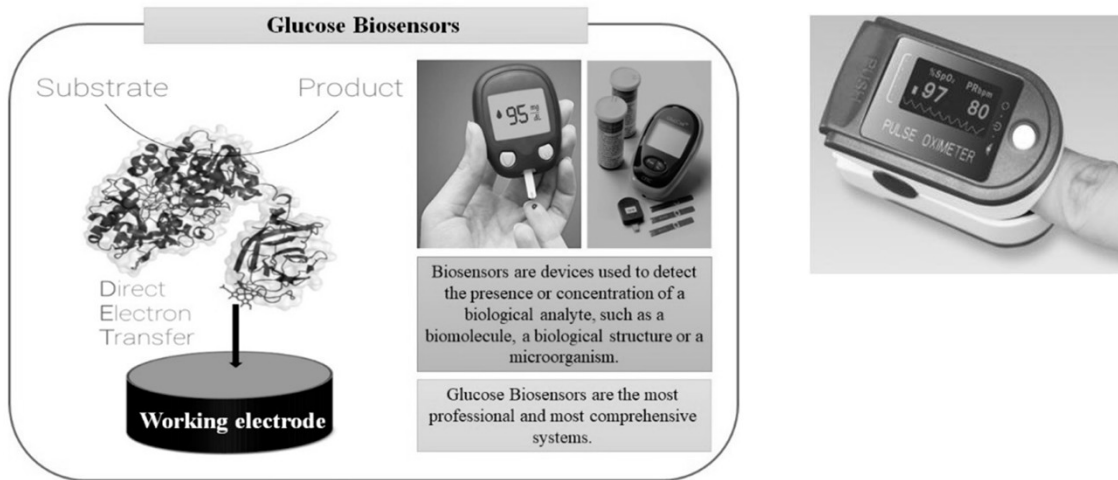
## Bio-sensors



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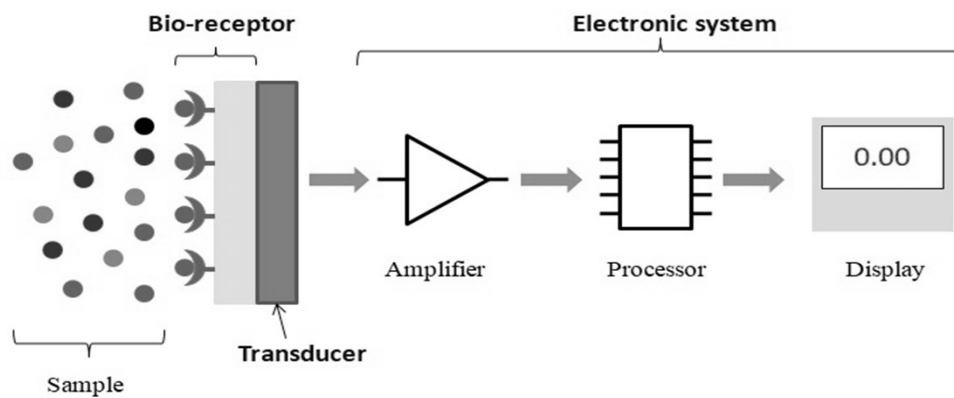


## Bio-sensors



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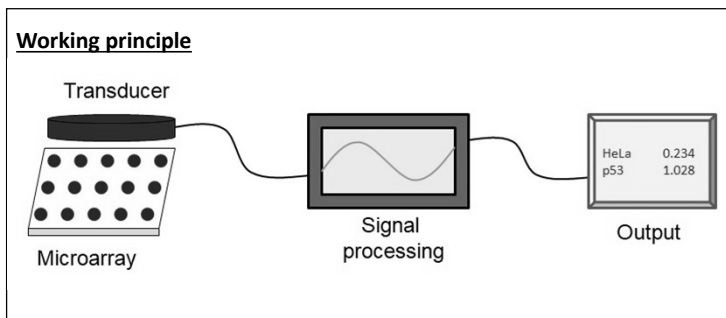
## Main components of Bio-sensor



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## Bio-chips

- Bio-chips are miniaturized laboratories in which thousands of biochemical reactions can be carried out simultaneously at micron scales for useful purposes such as disease studies or safety studies.
- Biochips are engineered substrates underlying sensor technology.



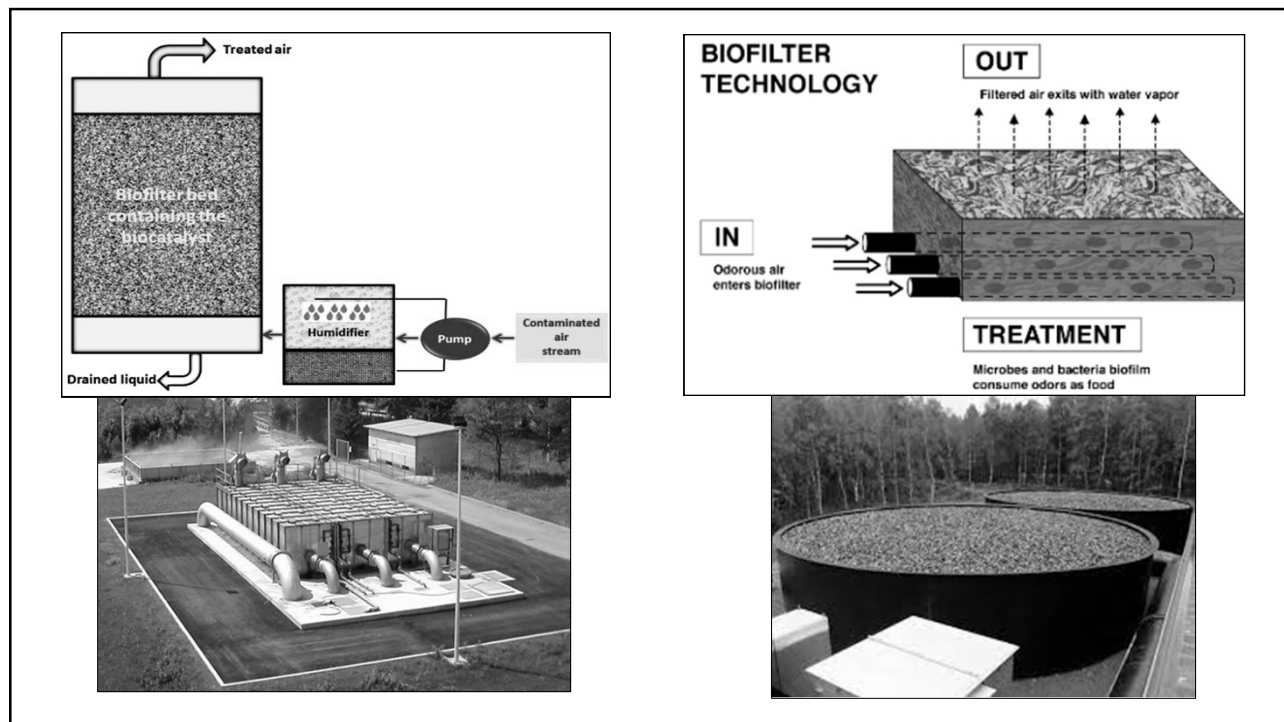
**Genechip** product contains thousands of individual DNA sensors use in sensing defect.

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## Bio-filters

- Biofilters are technical applications that use the biofiltration process to remove pollutants in environment.
- **Biofiltration** is a pollution control technique using a bioreactor containing living material to capture and biologically degrade pollutants.
- It is application are used to remove the pollutants from air and water.
- Some examples of biofiltration: Slow sand filters, Treatment ponds, Constructed wetlands and natural wetlands. Filters like Bioswales, biostrips, biobags, bioscrubbers, Vermifilters and trickling filters.

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## Bio-pesticides

- Bio-pesticides are biological substances or organisms that can be used instead of chemicals for pest control and thus they overcome the negative effects of chemical pesticides.
- The effective control of bio-pesticides requires appropriate formulation and application.
- Bio-pesticides are majorly used in the area of soil amendements and seed treatments.

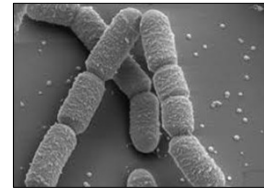


Micro-organism *Bacillus thuringiensis* stains use to kill insect larvae

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## Concrete Self-heal

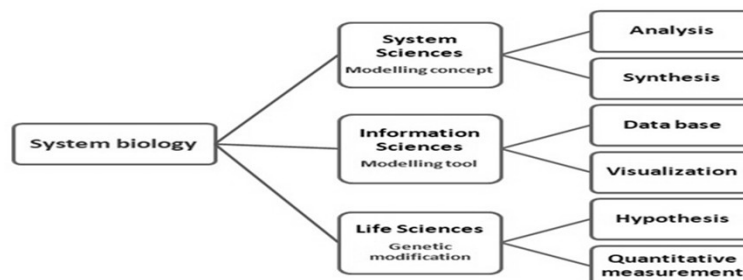
- Organisms can be used to make *concrete self-heal* its cracks due to wear-and-tear. For example, some bacteria (alkali-tolerant **bacteria** species such as *Bacillus pseudofirmus*) can catalyse the formation of calcium carbonate in their surroundings under appropriate conditions.
- When this happens in cracks that are formed in the concrete, the microscopic cracks are filled with the calcium carbonate formed with the help of the bacteria, which can effectively seal the cracks, and thus effect self-healing of the concrete.



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## System biology

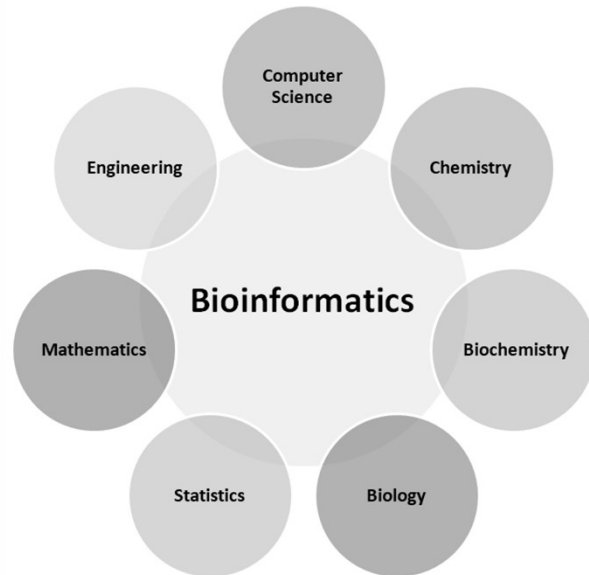
- Systems biology is the computational and mathematical analysis and modelling of complex biological systems.
- It is a biology-based interdisciplinary field of study that focuses on complex interactions within biological systems.
- Systems biology is a holistic approach in biomedical research to better understanding of complex picture of biology.



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## Bioinformatics

- These are currently popular fields of study which are highly multi-disciplinary, and engineers can significantly contribute to those fields.
- Those fields of study computationally analyze very large data sets to draw insights into the working of the fundamental functional unit of life - the cell.



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## Best Biomimicry Examples



**Inspired by:** Birds

**Where:** Italy & America

**Innovation / Function:** Aircraft

**Who:** Leonardo da Vinci & the Wright Brothers (shared credit)



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## Best Biomimicry Examples



**Inspired by:** Bur seeds of the burdock plant

**Where:** Switzerland

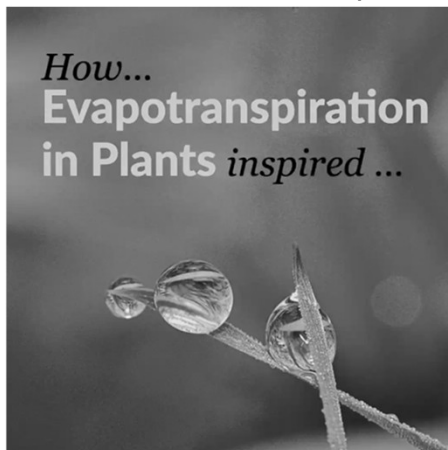
**Innovation/function:** Non-adhesive, easy to fix and remove clamping mechanism

**Who:** George de Mestral



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## Best Biomimicry Examples



**Inspired by:** Evapotranspiration in trees

**Where:** Massachusetts, USA

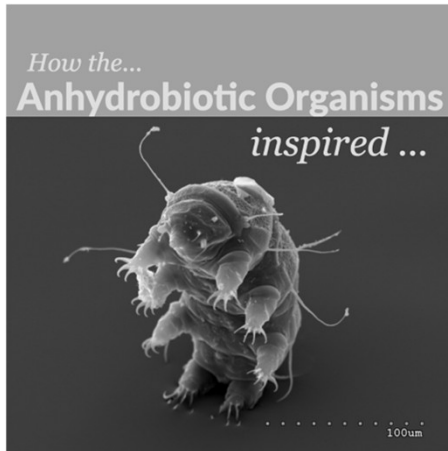
**Innovation / Function:** Waterless, low-cost, portable toilet

**Who:** change:WATER Labs



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## Best Biomimicry Examples

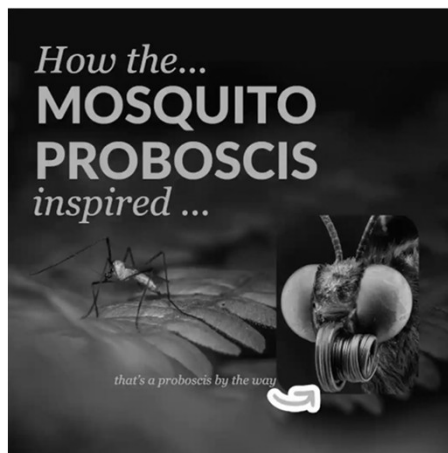


**Inspired by:** Resilient anhydrous organisms  
**Where:** Leicestershire, UK  
**Innovation / Function:** Long term vaccine storage  
**Who:** Nova Laboratories

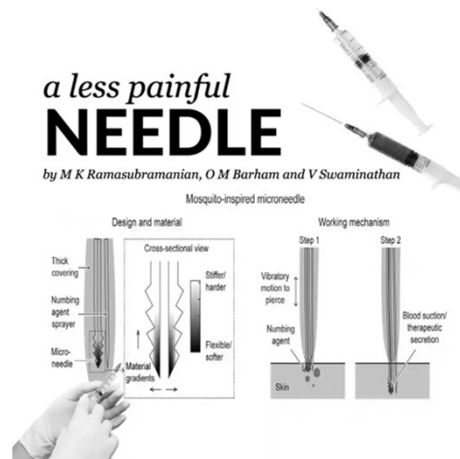


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## Best Biomimicry Examples



**Inspired by:** Mosquito proboscis  
**Where:** India & Japan  
**Innovation/function:** Less painful needle  
**Who:** M K Ramasubramanian, O M Barham and V Swaminathan



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## Best Biomimicry Examples



**Inspired by:** Sycamore seed pod  
**Where:** NSW, Australia  
**Innovation / Function:** Beautiful and efficient ceiling fan  
**Who:** Sycamore Technologies

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# THANK YOU

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